Status of the work in the folder Revision_Deflation_CJT_and_Kohn-Sham_definition_C

PROJECT spacetimes (author of the project: Patrick Lee Nash)

13 November 2026, 15:40 Pacific time

1. What this folder is

On 13 November 2026 the author of the project decided that, in a new folder, the self-interaction of the Dirac field on the $4+4$ spacetime is to be *defined* by the covariant form, that both forms are to be recorded correctly and completely, and that the two options are to be explained and discussed in the documents and in the provenance files. This is that folder. It is **work in progress**: this file states exactly what the folder contains at the time written above and what is still missing. The folder `Revision_Deflation_CJT_and_Kohn-Sham` of the same repository, in which the other form is the definition, is complete and is not changed by this work.

2. The two forms, in one page

The Dirac field ψ has sixteen components. The Dirac matrices γ^A $(A = 0, \dots, 7)$ of the $4+4$ spacetime are real 16×16 matrices; those of the four space-like coordinates $x0, \dots, x3$ square to $+1$, those of the four time-like coordinates $x4, \dots, x7$ to -1. Three matrices are built from them: $C^- = \gamma^0\gamma^1\gamma^2\gamma^3$ (real, symmetric, square 1), $\beta = -i\gamma^4$ (Hermitian, square 1), and $W = iC^-\gamma^4$ (Hermitian, square 1), so that $C^- = -W\beta$. For fields that do not depend on the three additional times $x5, x6, x7$ (the *quantized sector*) the kinetic and mass terms are taken in the positive form $\psi^\dagger(i\partial_4 - h)\psi$, with the Hamiltonian matrix $h = \sum_{j=0}^{3} \alpha^j(-i\partial_j) + \beta m$, $\alpha^j = -\gamma^4\gamma^j$; W commutes with h, and the field consists of two halves, $W = +1$ and $W = -1$.

The self-interaction is $\mathcal{L}_{\text{self}} = -\frac{1}{2}\sum_{r=0}^{8} g_r\, I_r$ with nine real couplings g_r and nine invariants I_r, one for each number r of different indices in a product of Dirac matrices. The two forms differ in the bilinears from which the invariants are built:

- **Form C (covariant form; the definition in this folder).** The bilinears are $\tilde{\psi}\,\Theta^K\psi$ with $\tilde{\psi} = \psi^\dagger C^-$ and $\Theta^K = \zeta_r\,\gamma^{A_1}\cdots\gamma^{A_r}$, where $\zeta_r = i$ for $r = 1, 2, 5, 6$ and $\zeta_r = 1$ otherwise; $I_r = \sum_{|K|=r}(\tilde{\psi}\,\Theta_K\psi)(\tilde{\psi}\,\Theta^K\psi)$ with $\Theta_K = (\Theta^K)^{-1}$. Each I_r is invariant under all transformations of $Spin(4,4)$.
- **Form Q (quantized-sector form; the alternative).** The bilinears are $\bar{\psi}\,\Theta^K\psi$ with $\bar{\psi} = \psi^\dagger\beta$ and with the matrices $-i\gamma^A$ $(A = 0, \dots, 4)$, γ^A $(A = 5, 6, 7)$ in place of γ^A, and $\zeta_r = i$ for $r = 2, 3, 6, 7$.

The exact relation between them (verified for all 256 index sets K): form C is form Q with the matrix W inserted in every bilinear, $I_r^{\text{C}} = \sum_{|K|=r}\epsilon_K\,(\psi^\dagger W\beta\,\Theta^K\psi)^2$ with the matrices Θ^K and the

signs $\epsilon_K = \pm 1$ of form Q. Two particles of the same half interact in the same way in both forms; two particles of different halves interact with opposite signs.

3. What the folder contains now

item	state
`code/deflation-cjt-ks-c-rs`	the program `deflation_cjt_ks_c` (Rust). It is the program of the folder with the other definition plus: the module `halves.rs` (uniform matter of the two halves of the field in both forms, the effective potential of Cornwall, Jackiw and Tomboulis for two halves, the vacuum with the filled states of negative energy, the functional with the Vorton-Scalar); the command `forms` (files `cmd_forms.rs`, `cmd_forms_b.rs`); extensions of `selfint.rs` (exact checks 44-S13 and S12 for all three models), of `pair_contact.rs` and `cmd_pairing.rs` (both forms in the pair along the hidden axis and in the pairs along $x5, x6, x7$), and of `cmd_gravity.rs` (matter of both forms as a source). It compiles; the unit tests of the new modules pass.
`code/scripts`	the scripts of the other folder with the names of this folder; not yet adapted to the new results.
`data`	only files that the program of this folder has written and whose command has finished are committed. At the time written above that is the command `superfluid`; the other commands are being executed.
documents (two manuscripts, the document on the Dirac matrices, the provenance files, README)	**not yet written for this folder**; they follow, each in the three formats .md, .tex and .pdf.

4. Exact results that are already established

They come from the command `selfcoupling` of the program of this folder (36 of 36 checks passed in a first run; integer arithmetic).

1. *The same four numbers in both forms.* For uniform matter whose two halves are filled differently, with the number densities $n_\pm$ and the scalar densities $s_\pm$ of the halves, the interaction energy is $\frac{1}{2}a_V(n_+^2 + n_-^2) + \frac{1}{2}a_S(s_+^2 + s_-^2) \pm [\, b_V\, n_+ n_- + b_S\, s_+ s_-]$, upper sign for form Q and lower sign for form C, with $a_S = (7g_{0\mathrm{eff}} - 5g_{1\mathrm{eff}} - 3g_{7\mathrm{eff}})/8$, $a_V = (-g_{0\mathrm{eff}} + 11g_{1\mathrm{eff}} - 3g_{7\mathrm{eff}})/8$, $b_S = (8g_{0\mathrm{eff}} - 3g_{1\mathrm{eff}} - 5g_{7\mathrm{eff}} - g_{8\mathrm{eff}})/8$, $b_V = (11g_{1\mathrm{eff}} - 3g_{7\mathrm{eff}} + g_{8\mathrm{eff}})/8$. Here $g_{0\mathrm{eff}}, g_{1\mathrm{eff}}, g_{7\mathrm{eff}}, g_{8\mathrm{eff}}$ are the four independent combinations of the nine couplings (scalar, vector, axial vector, pseudoscalar): $g_{0\mathrm{eff}} = g_0 - 3g_2 - 4g_3 - 5g_4 - 4g_5 - g_6$, $g_{1\mathrm{eff}} = g_1 - g_2 - g_5 + g_6$, $g_{7\mathrm{eff}} = g_7 - g_2 - g_3 + g_6$, $g_{8\mathrm{eff}} = g_8 - g_2 + 4g_3 - 5g_4 + 4g_5 - 3g_6$.

2. *The internal symmetry.* Fields that do not depend on $x0, x5, x6, x7$ have, without self-interaction, an internal symmetry with eight generators: the six products $\gamma^a\gamma^b$ of two of the matrices $\gamma^0, \gamma^5, \gamma^6, \gamma^7$, the unit matrix, and a matrix P_E proportional to $\gamma^0\gamma^5\gamma^6\gamma^7$. In form Q the self-interaction is invariant under the six products for all couplings, and under P_E if and only if $g_{1\text{eff}} = g_{7\text{eff}} = 0$. In form C it is invariant for all couplings only under the three products $\gamma^5\gamma^6, \gamma^5\gamma^7, \gamma^6\gamma^7$ (the rotations of the three additional times, which are symmetries of the spacetime); under the three unitary transformations generated by $\gamma^0\gamma^t$ ($t = 5, 6, 7$) and under P_E it is invariant only if all four effective couplings vanish. *In form C every self-interaction that does not vanish breaks the internal symmetry down to the rotations of* $x5, x6, x7$.
3. *Pairs.* The pair along the hidden axis, created by $\psi^T C^- \gamma^0 \psi$, consists of one particle of each half; the pairs along $x5, x6, x7$ consist of two particles of the same half. In form C the contact self-interaction therefore acts in the pair along the hidden axis with the sign opposite to that of form Q, and in the pairs along $x5, x6, x7$ with the same sign (unit test of `pair_contact.rs`, all momenta in ordinary space).

5. What is still missing

1. The data files of all commands, written by the program of this folder (in progress).
2. The independent verification of the new results by a second program in another language, and the test of that verification with deliberate errors.
3. The figures.
4. The documents: both manuscripts, the document on the Dirac matrices with both forms written out in full, the provenance files, among them one that records both options completely and discusses them, and the README; each as .md, .tex and .pdf.

Each of these items is committed and pushed when it is finished.